Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
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Project Title:	Impact of Microgravity on Drug and Nutrient Absorption		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-1065	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOte: * true end date is 10/31/2003		
Key Personnel Changes/Previous PI:	Amidon, Gordon		
COI Name (Institution):			
Grant/Contract No.:	NAG 9-1154		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term goal of the proposed research is to identify gastrointestinal physiological alterations which are likely to occur in space flight and which will lead to profound alterations in drug and nutrient absorption. This proposal has three Specific Aims. (1) To utilize our pellet gastric emptying test to evaluat alterations in gastric emptying during antiorthosatatic bed rest conditions (ABR). (2) Determine the alterations in intestinal transit using the lactulose test during the ABR conditions simultaneously with the PGE test (Specific Aim 1) in order to discriminate gastric emptying changes from intestinal transit changes. (3) Identify alterations in the intestinal membrane permeability Peff of several model drugs and nutrients to determine the intestinal phenotype for absorption. The aims of the proposed research will be achieved through a study of normal subjects (under 3-day ambulatory) as well as a 5-day bed rest period. The study will include gastric emptying and intestinal transit test prior to, during and following the ABR test. These studies will result in fundamental mechanistic information regarding gastrointestinal physiological changes that will impact the drug		

	and nutrient absorption. It will lead to a better understanding of how to correctly dose drugs in a micro-gravity environment during extended stays in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastric emptying, intestinal transit, membrane permeation and intestinal metabolism all impact drug absorption and systemic availability. This study will develop a test to determine when alterations in these processes impact drug systemic availability. It will be valuable in developing new drugs and drug delivery systems as well as understanding the effect of diet, disease and gravity on drug systemic availability.
Task Progress:	The long term objective of the proposed research was to identify gastrointestinal physiological alterations which may occur in space flight and may lead to alterations in drug and nutrient absorption. The following results were achieved during the period of this grant. Drug absorption models were developed based on cyclical motility patterns in the gastrointestinal tract that lead to variable gastric emptying, intestinal transit, and drug absorption to allow more accurate monitoring of drug absorption variation and the correlation with gastrointestinal motility (Time-dependent oral absorption models. Higaki K, Yamashita S, Amidon GL. J Pharmacokinet Pharmacodyn. 2001 Apr;28(2):109-28.). The results of this study are summarized in the abstract from the published paper on this work. “The plasma concentration-time profiles following oral administration of drugs are often irregular and cannot be interpreted easily with conventional models based on first- or zero-order absorption kinetics and lag time. Six new models were developed using a time-dependent absorption rate coefficient, ka(t), wherein the time dependency was varied to account for the dynamic processes such as changes in fluid absorption or secretion, in absorption surface area, and in motility with time, in the gastrointestinal tract. In the present study, the plasma concentration profiles of propranolol obtained in human subjects following oral dosing were analyzed using the newly derived models based on mass balance and compared with the conventional models. Nonlinear regression analysis indicated that the conventional compartment model including lag time (CLAG model) could not predict the rapid initial increase in plasma concentration after dosing and the predicted Cmax values were much lower than that observed. On the other hand, all models with the time-dependent absorption rate coefficient, ka(t), were superior to the CLAG model in predicting plasma concentration profiles. Based on Akaike's Information Criterion (AIC), the fluid absorption model without lag time (FA model) exhibited the best overall fit to the data. The two-phase model including lag time, TPLAG model was also found to be a good model judging from the values of sum of squares. This model also described the irregular profiles of plasma concentration with time and frequently predicted Cmax values satisfactorily. A comparison of the absorption rate profiles also suggested that the TPLAG model is better at prediction of irregular absorption kinetics than the FA model. In conclusion, the incorporation of a time-dependent absorption rate coefficient ka(t) allows the prediction of nonlinear absorption characteristics in a more reliable manner.” The oral absorption model developed in our laboratories has been applied to modeling a particularly complex absorption pattern observed for cimetidine which includes double peaks in the plasma level time curves (Variability in cimetidine absorption and plasma double peaks following oral administration in the fasted state in humans: correlation with antral gastric motility. Takamatsu N, Welage LS, Hayashi Y, Yamamoto R, Barnett JL, Shah VP, Lesko LJ, Ramachandran C, Amidon GL. Eur J Pharm Biopharm. 2002 Jan;53(1):37-47.) “The role of gastrointestinal motility and pH in determining cimetidine bioavailability as well as double peaks in plasma profiles following oral administration, in the quiescent or active phase of antral motility, to humans in the fasted state was examined. Plasma cimetidine-time curves did not show the presence of double peaks in any subject following intravenous administration. The incidence of double peaks was 73% following oral administration and was independent of antral migrating motility complex phase. Further, it was found that oral administration of cimetidine in the quiescent phase resulted in significantly higher bioavailability and in other pharmacokinetic parameters compared to that obtained following administration in the active phase. Excellent linearity in plots of motility peaks vs. plasma peaks with slopes close to unity were evident for both quiescent (r(2)=0.93) and active phase (r(2)=0.97) administration. A total of 14 peaks out of 22 (10 subjects, 64%) and 20 out of 27 peaks (11 subjects, 74%), were accounted for in quiescent and active phase oral administration, respectively. The proximal occurrence of plasma peaks to antral motility peaks typical of phase III contractions strongly implies that motility patterns may be responsible for secondary maxima following oral cimetidine administration in the fasted state.” We have further tested our pellet gastric emptying test to noninvasively measure changes in gastric emptying in the fasted-state in microgravity environment. This test is based on a salivary sampling and thus does not require blood withdrawal. The results of this validation study are summarized in the abstract below (Novel method to assess gastric emptying in humans: the Pellet Gastric Emptying Test. Choe SY, Neudeck BL, Welage LS, Amidon GE, Barnett JL, Amidon GL. Eur J Pharm Sci. 2001 Dec;14(4):347-53.). “To further validate the Pellet Gastric Emptying Test (PGET) as a marker of gastric emptying, a randomized, four-way crossover study was conducted with 12 healthy subjects. The study consisted of oral co-administration of enteric coated caffeine (CAFF) and acetaminophen (APAP) pellets in four treatment phases: Same Size (100 kcal), Fasted, Small Liquid Meal (100 kcal), and Standard Meal (847 kcal). The time of first appearance of measurable drug marker in plasma, t(initial), was taken as the emptying time for the markers. Co-administration of same size enteric coated pellets of CAFF and APAP (0.7 mm in diameter) revealed no statistically significant differences in t(initial) values indicating that emptying was dependent only on size and not on chemical make-up of the pellets. Co-administration of different size pellets indicated that the smaller 0.7-mm diameter (CAFF) pellets were emptied and absorbed significantly earlier than the larger 3.6-mm diameter (APAP) pellets with both the Small Liquid Meal (by 35 min) and the Standard Meal (by 33 min) ($P<0.05$). The differences in emptying of the pellets were not significant in the Fasted Phase. The results suggest that the pellet gastric emptying test could prove useful in monitoring changes in transit times in the fasted and fed states and their impact on drug absorption.” In a human study we have measured gene expression and correlated the gene expression with the absorption of a carrier-mediated antiviral drug, valacyclovir. The correlations observed in this study showed that the absorption of the carrier-mediated drug did indeed correlate with a transporter that has been reported to transport peptide compounds in vitro. The conclusions of this study are summarized in the abstract below (Gene expression in the human intestine and correlation with oral valacyclovir pharmacokinetic parameters. Landowski CP, Sun D, Foster DR, Menon SS, Barnett JL, Welage LS, Ramachandran C, Amidon GL. J Pharmacol Exp Ther. 2003 Aug;306(2):778-86. Epub 2003 May 15.) “The transport of valacyclovir, the l-valyl ester of acyclovir, has been suggested to be mediated by several carrier-mediated pathways in cell culture and animal models. The role and importance of these transporters in modulating valacyclovir absorption in humans has not been determined, however. Recent advances in genomic technology have facilitated the rapid and simultaneous determination of global mRNA expression profiles for thousands of genes in tissue biopsies directly associated with the absorption process, thereby dramatically increasing the value of

	studies in humans. In this article, we describe correlations of pharmacokinetic parameters following oral valacyclovir or acyclovir administration with expression levels of intestinal genes in humans. Highly positive and significant correlations were observed with 4F2hc, an activator of cation-preferring amino acid transport systems, and human oligopeptide transporter (HPT1), an oligopeptide transporter expressed at higher levels in the human intestine compared with oligopeptide transporter (PEPT1). The validation of HPT1 microarray data with reverse transcription-polymerase chain reaction and the enhanced valacyclovir uptake in HeLa/HPT1 cells suggest that the role of HPT1 in transport of peptides and peptidomimetics drugs needs to be examined in more detail. The interrelation of 4F2hc and HPT1 in transport may be of interest. No significant correlations of valacyclovir pharmacokinetic parameters with PEPT1 and with organic cation or anion transporter expression levels were observed. The highly negative correlations observed with known efflux pumps such as MDR1 (P-glycoprotein) and MRP2 (cMOAT), as well as with the CYP450 IIIA subfamily may indicate that these proteins may regulate the cellular accumulation and metabolism of acyclovir.”
Bibliography Type:	Description: (Last Updated: 03/19/2008)
Articles in Peer-reviewed Journals	Landowski CP, Sun D, Foster DR, Menon SS, Barnett JL, Welage LS, Ramachandran C, Amidon GL. "Gene expression in the human intestine and correlation with oral valacyclovir pharmacokinetic parameters." J Pharmacol Exp Ther. 2003 Aug;306(2):778-86. PMID: 12750437 , Aug-2003
Articles in Peer-reviewed Journals	Takamatsu N, Welage LS, Hayashi Y, Yamamoto R, Barnett JL, Shah VP, Lesko LJ, Ramachandran C, Amidon GL. "Variability in cimetidine absorption and plasma double peaks following oral administration in the fasted state in humans: correlation with antral gastric motility." Eur J Pharm Biopharm. 2002 Jan;53(1):37-47. Erratum in: Eur J Pharm Biopharm. 2002 Sep;54(2):255. PMID: 11777751 , Jan-2003
Articles in Peer-reviewed Journals	Choe SY, Neudeck BL, Welage LS, Amidon GE, Barnett JL, Amidon GL. "Novel method to assess gastric emptying in humans: the Pellet Gastric Emptying Test." Eur J Pharm Sci. 2001 Dec;14(4):347-53. PMID: 11684410 , Jan-2003
Articles in Peer-reviewed Journals	Higaki K, Yamashita S, Amidon GL. "Time-dependent oral absorption models." J Pharmacokinet Pharmacodyn. 2001 Apr;28(2):109-28. PMID: 11381566 , Jan-2003